Hints: Euclidean geometry – 1. Vectors, angles, simplices and volumes in $\mathbb{R}^n$

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Key lemma: non-zero vectors with pairwise non-acute angles that all lie in one open half-space $\{\langle x, w\rangle > 0\}$ are linearly independent. Indeed, split a dependence $\sum c_i v_i = 0$ by the signs of c_i: $x = \sum_{c_i>0} c_i v_i = \sum_{c_j<0} |c_j| v_j$, so $|x|^2 = \sum c_i |c_j| \langle v_i, v_j\rangle \leqslant 0$ and $x = 0$. But $\langle x, w\rangle > 0$ if some $c_i > 0$; hence no c_i is positive, and by the same argument for $-c$ none is negative.

Hint 2. Bound $2n$: take w orthogonal to none of the v_i; the vectors with $\langle v_i, w\rangle > 0$ and those with $\langle v_i, w\rangle < 0$ are independent, so there are at most n of each. Bound $n+1$: take $w = -v_m$; then $v_1, \dots, v_{m-1}$ lie in the half-space, so $m - 1 \leqslant n$. Examples: $\pm e_1, \dots, \pm e_n$, and the vectors from the centre of a regular simplex to its vertices.

Hint 3. Structure for $m = 2n$: project onto $v_m^\perp$. The projections are still pairwise non-acute ($\langle w_i, w_j\rangle = \langle v_i, v_j\rangle - \frac{\langle v_i, v_m\rangle\langle v_j, v_m\rangle}{|v_m|^2} \leqslant \langle v_i, v_j\rangle$), so at most $2n-2$ of them are non-zero: some $v_k = -c v_m$, $c > 0$. For any other v_i both $\langle v_i, v_m\rangle \leqslant 0$ and $\langle v_i, v_k\rangle = -c\langle v_i, v_m\rangle \leqslant 0$, so $v_i \perp v_m$. Induct in $v_m^\perp$.

Hint 4. Last claim: $0 \leqslant \left|\sum v_i\right|^2 = m + \sum_{i\neq j}\langle v_i, v_j\rangle \leqslant m - m(m-1)\beta$.

Problem 2

Hint 1. Start with (b). Fix a unit vector u and project S onto $u^\perp$. The facets with $\langle u_i, u\rangle > 0$ cover the projection of S once, and so do those with $\langle u_i, u\rangle < 0$; by Theorem 2 the facet opposite A_i contributes $F_i|\langle u_i, u\rangle|$. Hence $\sum_i F_i\langle u_i, u\rangle = 0$ for every u. (This is the divergence theorem for a constant vector field.)

Hint 2. From $F_0 u_0 = -\sum_{i\geqslant 1} F_i u_i$: taking lengths gives $F_0 < \sum_{i\geqslant 1} F_i$ (the u_i are not all parallel); squaring gives the cosine formula, since the outer normals of two facets make the angle $\pi - \varphi_{ij}$.

Hint 3. (a) is now one line: if the edges at A_0 are orthogonal, the facets opposite $A_1, \dots, A_n$ lie in the coordinate hyperplanes through A_0, so $u_i = -e_i$ and $F_0^2 = \left|\sum_{i\geqslant 1} F_i e_i\right|^2 = \sum_{i\geqslant 1} F_i^2$.

Problem 3

Hint 1. Put $v_i = A_i - B$. Since B is interior, $\sum \lambda_i v_i = 0$ with all $\lambda_i > 0$ (barycentric coordinates). Consider the graph on $\{1, \dots, n+1\}$ in which i and j are joined when $\langle v_i, v_j\rangle < 0$, i.e. when $\angle A_i B A_j > 90^\circ$. It suffices to show that this graph is connected.

Hint 2. If not, split the indices into two non-empty parts I, J with $\langle v_i, v_j\rangle \geqslant 0$ for $i \in I$, $j \in J$. Then $w = \sum_{i\in I}\lambda_i v_i = -\sum_{j\in J}\lambda_j v_j$ and $|w|^2 = -\sum_{i\in I, j\in J}\lambda_i\lambda_j\langle v_i, v_j\rangle \leqslant 0$ (the lemma of Theorem 3).

Hint 3. So $\sum_{i\in I}\lambda_i v_i = 0$: B is a convex combination of the vertices A_i, $i \in I$, i.e. it lies on a proper face of the simplex, not inside. A connected graph on $n+1$ vertices has at least n edges.

Problem 4

Hint 1. Work in barycentric coordinates: $M = \sum \lambda_j A_j$ with $\lambda_j > 0$, $\sum \lambda_j = 1$. The point N_i is on the line $A_i M$ and has i-th coordinate 0, so $N_i = \frac{M - \lambda_i A_i}{1-\lambda_i}$, with coordinates $\frac{\lambda_j}{1-\lambda_i}$ ($j \neq i$) and 0 at i.

Hint 2. The ratio of volumes is the absolute value of the determinant of the matrix of barycentric coordinates of $N_1, \dots, N_{n+1}$. This matrix is $D^{-1}(\mathbf{1}\lambda^T - \Lambda)$ with $D = \operatorname{diag}(1-\lambda_i)$, $\Lambda = \operatorname{diag}(\lambda_i)$, and $\det(\mathbf{1}\lambda^T - \Lambda) = \det(-\Lambda)\big(1 - \lambda^T\Lambda^{-1}\mathbf{1}\big)$. So the ratio equals

$$\frac{n\,\lambda_1 \cdots \lambda_{n+1}}{(1-\lambda_1)\cdots(1-\lambda_{n+1})}.$$

Hint 3. By AM–GM, $1-\lambda_i = \sum_{j\neq i}\lambda_j \geqslant n\big(\prod_{j\neq i}\lambda_j\big)^{1/n}$; multiplying over i gives $\prod(1-\lambda_i) \geqslant n^{n+1}\prod\lambda_j$. Hence the ratio is at most $\frac{1}{n^n}$, with equality exactly for $\lambda_i = \frac{1}{n+1}$: M is the centroid.

Problem 5

Hint 1. A description of the circumcentre: let L_T be the affine function equal to $|x|^2$ at the vertices of the simplex T. Since $|x-C|^2 - R^2 = |x|^2 - 2\langle C, x\rangle + \text{const}$ vanishes at the vertices, $L_T(x) = 2\langle C(T), x\rangle + \text{const}$, i.e. $C(T) = \frac{1}{2}\nabla L_T$. So we need $\sum_i v(S_i)\nabla L_{S_i} = v(S)\nabla L_S$.

Hint 2. Let f be the function equal to L_{S_i} on each S_i. It is continuous (on a common facet both are the affine interpolation of $|x|^2$ at the vertices of that facet), and on ∂S it coincides with L_S (every facet of S is a facet of one S_i, with the same vertices).

Hint 3. $\sum_i v(S_i)\nabla L_{S_i} = \int_S \nabla f$. For a continuous piecewise affine f, $\int_S \frac{\partial f}{\partial x_k}$ depends only on $f|_{\partial S}$: on every line parallel to e_k the integral is $f(\text{exit}) - f(\text{entry})$. Replacing f by L_S gives $v(S)\nabla L_S$.